

Time: 90 minutes

Available marks: 62

Full marks: 60

Instructions

- 1. Fill in your details in the space provided below. You may also enter your student number on the top of each page in the space provided.
- 2. If you have read the instructions, write the name of your favourite mathematician on the back of this page. You will get one extra mark if you include a reason why they are your favourite.
- 3. Answer all questions on this question paper in the boxes provided. You may use the backs of pages if you run out of space.
- 4. Write neatly and legibly, otherwise your answer will not be marked.
- 5. Show all work and reasoning unless otherwise instructed.
- 6. This test consists of 4 questions and 5 numbered pages. Make sure your question paper is complete.

Good luck! ✨⁺

Student Number

--	--	--	--	--	--	--	--	--

Name & Last Name

--	--	--	--	--	--	--	--	--

1. Let G be a group. We define $*$: $(G \times G) \times G \rightarrow G$ by $(g, h) * x = gxh^{-1}$.

(a) Prove that $*$ is a group action by $G \times G$ on G .

(b) Prove that all elements of G belong to the same orbit.

(c) Find $S(1)$.



2. (a) Prove the following lemma:

Lemma. *If a finite nontrivial group has no nontrivial proper subgroups, then it has prime order.*

- (b) Let G be a finite group with exactly three subgroups. Prove that G has order p^2 for some prime p .

HINT: Examine the nontrivial proper subgroup of G . What is its order? Is it normal? What would the order of its factor group be? Make use of the lemma.

--	--	--	--	--	--	--	--	--

3. (a) Let G be a p -group and $*$ be a group action by G on a finite set X .

(i) State the definition of X_f .

(ii) Write a statement that relates $|X|$ to $|X_f|$, as seen in class.

--

(b) Let G be a p -group and H a proper subgroup of G .

(i) State the definition of $N_G(H)$.

(ii) Prove that $N_G(H) \neq H$. HINT: Let H act on the set of left cosets of H in G through left multiplication (you do not need to prove this is a group action).

--

--	--	--	--	--	--	--	--	--

4. (a) State the Third Sylow Theorem.

--

(b) Let G be a group of order $217 = 7 \times 31$. Prove or disprove: G is cyclic.

HINT: Determine if there are any elements of order 217.

--

[3 + 10 = 13 marks]